

Engineering Assessment — Katharina

Engineering Assessment — Freeze Snapshot

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Overall: Strong mid-to-senior systems+ML ops engineer. Clear safety mindset, fast surge debugging, and excellent product/clinical empathy. ML integration kept advice-only bou

Strengths

Standout strengths

- Contract discipline & safety rails: froze CONFIG; used overlay CFG for tests; guarded top-level side effects.
- Rapid, surgical debugging: fixed EquipmentRepository.read None-path with an idempotent patch; created tracker_core alias shim.
- Verification mindset: authored smokes; removed brittle checks; introduced an IO doctest.
- Product/clinical empathy: avoided double notifications; reframed Gate_pos into clinically meaningful targets.
- Pragmatic UI engineering: split Streamlit monolith; added adjustable vitals thresholds; optional deps lazy.
- Ops-ready ML: reused MLP advice-only; handled calibration safely; deterministic outputs; acknowledged version warnings.
- Documentation & handoff: produced Handoff/Summary PDFs and ChatGPT handoff with context; safe.

Level-up Opportunities

Level-up opportunities (high impact)

- Tests > smokes: add 2–3 pytest (repo read, QR fallback, SOP offline).
- Type rigor: expand type hints and run mypy on core repos/services.
- Metrics & calibration: track per-task Brier/ECE/PR-AUC; calibrate each head in multi
- Packaging: extract tracker/UI helpers into a tiny ed_tracker/ package; notebook stay
- Data contracts: define a minimal CSV schema object to prevent drift.

Skill Snapshot

- Skill snapshot (1–5)
- Systems design / safety rails: 5
 - Debugging & incident response: 5
 - DevEx & reproducibility: 4.5
 - Product/clinical reasoning: 4.5
 - ML integration (calibration, targets): 4
 - Testability/packaging hygiene: 3.5–4

Why This Fits Your Bio

Why this matches your bio

You described being good at debugging and newer to writing code. Your debugging is clear and your structuring (guards, overlays, helper splits, alias shim, robust IO) shows you're 'new' and into reliable, maintainable engineering. Your concern for confounding and eq workflows reflects active AI ethics thinking.

Next Steps

Next steps (when ready)

- Phase 2.5-3: build labels + multi-task heads (advice-only), calibrate, integrate as (never veto).
- Add SOP usage telemetry; optional: persist UI thresholds across sessions.